

DOCUMENTATION NAT/PAT

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PREMIÈRE ÉTAPE

DEUXIÈME ÉTAPE

Routeur A connecté au Routeur B

```
Routeur-A(config)#ip route 0.0.0.0 0.0.0.0 20.6.6.1
Routeur-A(config)#

Gateway of last resort is 20.6.6.1 to network 0.0.0.0

    20.0.0.0/30 is subnetted, 1 subnets
C       20.6.6.0 is directly connected, FastEthernet0/0
C       192.168.0.0/24 is directly connected, FastEthernet0/1
S*     0.0.0.0/0 [1/0] via 20.6.6.1
```

Mettre en place les routes par défaut :

→ Routeur(config)# ip route 0.0.0.0 0.0.0.0 adresse du prochain saut

Routeur F connecté au Routeur F

```
Routeur-F>en
Routeur-F#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Routeur-F(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2

    20.0.0.0/30 is subnetted, 1 subnets
C       20.0.0.0 is directly connected, FastEthernet0/1
C       192.168.0.0/24 is directly connected, FastEthernet0/0
S*     0.0.0.0/0 [1/0] via 20.0.0.2
```

TROISIÈME ÉTAPE

Mettre en place le protocole NAT sur les routeurs terminaux RtA et RtF afin que les postes clients W / Z puissent se joindre.

Routeur A

```
Routeur-A(config)#interface FastEthernet0/0
Routeur-A(config-if)#ip address 20.6.6.2 255.255.255.252
Routeur-A(config-if)#ip nat outside

interface FastEthernet0/0
 ip address 20.6.6.2 255.255.255.252
 ip nat outside
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 192.168.0.254 255.255.255.0
 ip nat inside
 duplex auto
 speed auto
```

Routeur F

```
interface FastEthernet0/0
 ip address 192.168.0.254 255.255.255.0
 ip nat inside
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 20.0.0.1 255.255.255.252
 ip nat outside
 duplex auto
 speed auto
```

ACCESS-LIST Routeur A

```
!
access-list 1 permit 192.168.0.0 0.0.0.255
```

ACCESS-LIST Routeur F

```
access-list 1 permit 192.0.0.0 0.255.255.255
```

ACTIVATION DU NAT RTA

```
access-list 1 permit 192.0.0.0 0.255.255.255
```

ACTIVATION DU NAT RTF

```
ip nat inside source list 1 interface FastEthernet0/1 overload
```

QUATRIÈME ÉTAPE

Mettre en place le routage dynamique et le protocole RIP : RtB, RtC, RtD et RtE

Les commandes à appliquer :

```
Routeur-B(config)#  
Routeur-B(config)#router RIP  
Routeur-B(config-router)#version 2  
Routeur-B(config-router)#NETWORK 172.11.0.1  
Routeur-B(config-router)#NETWORK 172.11.0.0  
Routeur-B(config-router)#NETWORK 20.6.6.0  
Routeur-B(config-router)#NETWORK 20.5.5.0  
Routeur-B(config-router)#NETWORK 20.4.4.0  
Routeur-B(config-router)#No Auto-Summary
```

Routeur B :

```
20.0.0.0/30 is subnetted, 4 subnets  
C      20.2.2.0 is directly connected, FastEthernet0/1  
C      20.4.4.0 is directly connected, Ethernet1/1  
C      20.5.5.0 is directly connected, Ethernet1/0  
C      20.6.6.0 is directly connected, FastEthernet0/0  
172.11.0.0/24 is subnetted, 1 subnets  
C      172.11.0.0 is directly connected, Ethernet1/2
```

Routeur C :

20.0.0.0/30 is subnetted, 5 subnets

```
C      20.1.1.0 is directly connected, FastEthernet0/0
R      20.2.2.0 [120/1] via 20.5.5.2, 00:00:14, FastEthernet0/1
R      20.4.4.0 [120/1] via 20.5.5.2, 00:00:14, FastEthernet0/1
C      20.5.5.0 is directly connected, FastEthernet0/1
R      20.6.6.0 [120/1] via 20.5.5.2, 00:00:14, FastEthernet0/1
172.11.0.0/24 is subnetted, 1 subnets
R      172.11.0.0 [120/1] via 20.5.5.2, 00:00:14, FastEthernet0/1
```

Routeur E :

20.0.0.0/30 is subnetted, 7 subnets

```
C      20.0.0.0 is directly connected, FastEthernet0/0
C      20.1.1.0 is directly connected, Ethernet1/0
C      20.2.2.0 is directly connected, FastEthernet0/1
C      20.3.3.0 is directly connected, Ethernet1/1
R      20.4.4.0 [120/1] via 20.2.2.2, 00:00:17, FastEthernet0/1
          [120/1] via 20.3.3.2, 00:00:25, Ethernet1/1
R      20.5.5.0 [120/1] via 20.2.2.2, 00:00:17, FastEthernet0/1
          [120/1] via 20.1.1.2, 00:00:00, Ethernet1/0
R      20.6.6.0 [120/1] via 20.2.2.2, 00:00:17, FastEthernet0/1
172.11.0.0/16 is variably subnetted, 2 subnets, 2 masks
R      172.11.0.0/16 is possibly down, routing via 20.3.3.2, Ethernet1/1
R      172.11.0.0/24 [120/1] via 20.2.2.2, 00:00:17, FastEthernet0/1
172.12.0.0/16 is variably subnetted, 2 subnets, 2 masks
R      172.12.0.0/16 is possibly down, routing via 20.1.1.2, Ethernet1/0
C      172.12.0.0/24 is directly connected, Ethernet1/2
```

Routeur D : récupère toutes les adresses IP configurées en amont sur les précédents routeurs.

```
20.0.0.0/30 is subnetted, 7 subnets
R    20.0.0.0 [120/1] via 20.3.3.1, 00:00:23, FastEthernet0/0
R    20.1.1.0 [120/1] via 20.3.3.1, 00:00:23, FastEthernet0/0
R    20.2.2.0 [120/1] via 20.4.4.2, 00:00:02, FastEthernet0/1
      [120/1] via 20.3.3.1, 00:00:23, FastEthernet0/0
C    20.3.3.0 is directly connected, FastEthernet0/0
C    20.4.4.0 is directly connected, FastEthernet0/1
R    20.5.5.0 [120/1] via 20.4.4.2, 00:00:02, FastEthernet0/1
R    20.6.6.0 [120/1] via 20.4.4.2, 00:00:02, FastEthernet0/1
172.11.0.0/16 is variably subnetted, 2 subnets, 2 masks
R    172.11.0.0/16 is possibly down, routing via 20.4.4.2, FastEthernet0/1
R    172.11.0.0/24 [120/1] via 20.4.4.2, 00:00:02, FastEthernet0/1
172.12.0.0/16 is variably subnetted, 2 subnets, 2 masks
R    172.12.0.0/16 is possibly down, routing via 20.3.3.1, FastEthernet0/0
R    172.12.0.0/24 [120/1] via 20.3.3.1, 00:00:23, FastEthernet0/0
```

CINQUIÈME ÉTAPE

Le protocole RiP fonctionne bien, les requêtes se transmettent de machine en machine pour arriver à sa destination.

```
C:\>ping 172.11.0.1

Pinging 172.11.0.1 with 32 bytes of data:

Reply from 172.11.0.1: bytes=32 time=6ms TTL=126
Reply from 172.11.0.1: bytes=32 time=6ms TTL=126
Reply from 172.11.0.1: bytes=32 time=35ms TTL=126
Reply from 172.11.0.1: bytes=32 time=2ms TTL=126

Ping statistics for 172.11.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 35ms, Average = 12ms

C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=1ms TTL=252
Reply from 20.0.0.2: bytes=32 time=7ms TTL=252
Reply from 20.0.0.2: bytes=32 time=1ms TTL=252
Reply from 20.0.0.2: bytes=32 time=1ms TTL=252

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 7ms, Average = 2ms
```

Voici un extrait :

On vérifie que le protocole RiP fonctionne bien, avec des routes supprimées

```
C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.0.254
  2  0 ms      0 ms      0 ms      20.6.6.1
  3  0 ms      0 ms      0 ms      20.5.5.1
  4  0 ms      1 ms      11 ms     20.0.0.2

Trace complete.

C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.0.254
  2  0 ms      0 ms      0 ms      20.6.6.1
  3  0 ms      0 ms      0 ms      20.0.0.2

Trace complete.
```

Version traçable du routeur :

```
Routeur-B>tracert 20.0.0.1
Type escape sequence to abort.
Tracing the route to 20.0.0.1

  1  20.2.2.1          0 msec    0 msec    0 msec
  2  20.0.0.1          0 msec    1 msec    0 msec
 _  _  |
```

SIXIÈME ÉTAPE

7.

Création de la route static avec le protocole TCP (80)

Afin de pouvoir communiquer avec le réseau du routeur A au routeur F.

```
-----  
Enter configuration commands, one per line.  End with CNTL/Z.  
Routeur-F(config)#ip nat inside source static tcp 192.168.0.200 80 20.0.0.1 80  
Routeur-F(config)#end  
Routeur-F#  
%SYS-5-CONFIG_I: Configured from console by console
```

Point final : l'ordinateur W peut communiquer sans problème avec le serveur Z

